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give me the control structures in python

ChatGPT said:

ChatGPT

In Python, control structures help you manage the flow of your program. Here are the main types:

1. Conditional Statements

These allow you to execute code based on certain conditions.

- **if statement**

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if condition:

code to execute if condition is True

- **if-else statement**

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if condition:

code if condition is True

else:

code if condition is False

- **if-elif-else statement**

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if condition1:

code if condition1 is True

elif condition2:

code if condition2 is True

else:

code if both conditions are False

2. Looping Statements

These allow you to repeat a block of code multiple times.

- **for loop**

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for item in iterable:

code to execute for each item

- **while loop**

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while condition:

code to execute as long as condition is True

- **break statement:** Exits the loop prematurely.

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for item in iterable:

if condition:

break

- **continue statement:** Skips the rest of the current loop iteration and moves to the next one.

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for item in iterable:

if condition:

continue

code that runs only if condition is False

3. Exception Handling

These structures allow you to manage errors and exceptions in your code.

- **try-except block**

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try:

code that may raise an exception

except SomeException:

code to execute if the exception occurs

- **finally block:** Executes code regardless of whether an exception occurred.

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try:

code that may raise an exception

except SomeException:

handle the exception

finally:

code that runs no matter what

4. Comprehensions

These provide a concise way to create lists, dictionaries, or sets.

- **List comprehension**

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new_list = [expression for item in iterable if condition]

- **Dictionary comprehension**

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new_dict = {key_expression: value_expression for item in iterable if condition}

Summary

These control structures help you make decisions, repeat tasks, handle errors, and create collections in a structured way. If you have specific scenarios in mind, I can provide more tailored examples!

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